

# Do Tri Nhan

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## TECHNICAL SKILLS

**Speech & Audio:** TTS, Speaker Verification & Diarization, Voice Cloning & Conversion, Anti-spoofing, Speech Enhancement, Sound Event Detection

**ML & LLM:** PyTorch, ONNX, Triton, TorchServe, SLM, Reinforcement Learning, Agno Agent, RAG, Speech SSL

**Engineering:** Docker, FastAPI, Flask, RabbitMQ, Milvus, Qdrant Vector DB, Slurm, Datadog, Jenkins, Jfrog

**Languages:** Python (primary), C++, Java, React Vibe

## EDUCATION

### VNUHCM – University of Science

Bachelor of Science, Advanced Program in Computer Science (APCS K17)

Sep 2017 – Jun 2021

Ho Chi Minh City, Vietnam

- GPA: 3.83 / 4.0    TOEFL ITP: 550 / 677

## EXPERIENCE

### Vinfast Automotive AI Development Institute – Vingroup JSC

Senior AI Engineer – Conversational Speech Technology

Jun 2026 – Present

Ho Chi Minh City, Vietnam

- **Unified TTS Serving Gateway:** Architected a production FastAPI gateway consolidating multiple TTS model tiers and normalization backends — supporting streaming, async job queue, API key auth with per-key usage tracking, and horizontal shard deployment for multilingual production serving.
- **TTS Normalization Automation Harness:** Designed an end-to-end QC automation framework with LLM-based error classification, BA-configurable per-project switch-code vocabulary management, automated Java SDK and prompt rebuild, and full regression evaluation — reducing engineering intervention for vocabulary updates to zero.

### VinSmart Future – Vingroup JSC

AI Team Lead – Speech Generation

Feb 2026 – Jun 2026

Ho Chi Minh City, Vietnam

- **Research Team Leadership:** Lead a team of 6 members on speech generation: voice design, non-verbal audio, and reinforcement learning for speech synthesis.
- **Voice Model Product Management:** Coordinate between research team and product projects; standardize quality assurance and release processes for voice models across three product lines: robot assistants, in-vehicle AI, and super-app voice mode.

Senior AI Engineer – Core AI Research

Jul 2025 – Feb 2026

- **Vietnamese Phoneme-Based Flow Matching TTS:** Developed a Vietnamese phoneme-adaptive fine-tuning TTS model achieving stable quality with low WER in production.
- **TensorRT & Triton Serving Optimization:** Optimized model serving with TensorRT and Triton Inference Server, achieving 2× throughput improvement and increased concurrent user capacity.
- **Hybrid Text Normalization:** Developed a Small Language Model (SLM) combined with rule-based normalization in a hybrid pipeline, improving accuracy by 13% over the rule-based baseline.

### Vingroup Big Data Institute – Vingroup JSC

Middle AI Engineer – AI Agent & Voice Biometric

Jan 2024 – Jul 2025

Hanoi, Vietnam

- **AI Agent with Knowledge Base:** Built a RAG-based AI agent with query processing, embedding, function calling, and automatic double-bot quality control evaluation (2025).
- **Speech Data Warehouse:** Managed a team building a speech data warehouse for storage, querying, processing, release management, and natural language querying integration - Data Intelligence (2025).
- **Streaming Diarization Optimization:** Optimized and served streaming diarization with Triton EEND-VC, achieving 2× speed and superior quality over third-party solutions (2025).
- **Anti-spoofing Module:** Researched and deployed an anti-spoofing system with an EER of 3.75%, combining RawBoost augmentation, speech SSL, and Graph Neural Networks (2024).

AI Engineer – Speech Synthesis & Voice Biometric

Oct 2021 – Dec 2023

- **Smart AI Voice Recording Service:** Designed an AI voice recording service for insurance operations using Speaker Diarization, ASR, and Voice Biometrics to detect agent fraud (2023).
- **Government Voice Biometric Project:** Implemented QC workflows and a microservice gateway contributing SNR Estimator, VAD, ASR, and Speaker Counter via TorchServe and Milvus (2023).
- **Multispeaker Acoustic Model:** Reduced GPU usage 4× via a multispeaker acoustic model supporting 30-min fine-tuning and zero-shot cloning with 30 s of audio (2023).
- **Universal Multistream Vocoder:** Optimized the EV assistant cloud TTS vocoder with Universal Multistream HiFi-GAN, achieving 1.5× faster inference without quality loss (2022).
- **AI Service User Interfaces:** Developed UI tools for AI services to support demos, data labeling, model evaluation, POC, and project bidding using Streamlit, Material UI React, and Wavesurfer.js (2022).

- **Tacotron2 Enhancement:** Enhanced robustness of Tacotron2 by integrating Monotonic Alignment Attention, improving phoneme–speech alignment and eliminating noise artifacts (2021).
- **Massive Speech Data Crawler:** Designed a large-scale automated speech data crawling and processing system covering data selection, formatting, denoising, transcription, quality ranking, and characteristic analysis (2021).

### VinAI Research Institute – Vingroup JSC

Engineering Resident – ERP Batch 1

Dec 2020 – Sep 2021

Hanoi, Vietnam

- **Vietnamese TTS for Electric Vehicles:** Integrated state-of-the-art TTS (Tacotron2, FastSpeech2, GlowTTS, HiFi-GAN), optimized with ONNX Runtime for edge inference.
- **SpeechMT:** Designed and deployed a real-time voice-based machine translation service containerized with Docker and served via FastAPI.
- **Speech-Based QA for Car Manuals:** Developed a lightweight speech-interactive system for voice-driven car manual queries, deployed end-to-end.

### KMS Technology

Software Engineer – Center of Excellence

Jun 2020 – Sep 2020

Ho Chi Minh City, Vietnam

- **RASA Chatbot Core Research:** Improved intent classification by combining Dense (ConveRT, BERT) and Sparse (TF-IDF, Count Vectors) features; evaluated on SQuAD data.
- **Reading Comprehension Backend:** Built BIDAf-based reading comprehension services with AllenNLP, Flask, Swagger, Docker Compose, and RabbitMQ.

### Scooter Saigon Tour – Tourism Startup

SEO Developer – Founder

Jan 2014 – Aug 2017

Ho Chi Minh City, Vietnam

- **Website Development:** Built a WordPress website with PHP backend and integrated VTCpay payment system.
- **SEO & Online Marketing:** Optimized site content, built backlink networks, and managed social media channels to improve search rankings.

### Freelance AI Projects (selected):

- Voice-Preserving Speech MT
- Singing Voice Conversion
- Pronunciation Assessment
- Speech Enhancement & Noise Reduction
- Sleep Stage Classification
- Emotional Dubbing

### RESEARCH AFFILIATIONS

#### Speech and Language Lab, Nanyang Technological University

Research Intern (Remote)

Sep 2022 – Mar 2023

Singapore

- Participated in a government project with ST Engineering, developing and evaluating a model for classifying emergency sound events in urban areas.

#### Artificial Intelligence Lab, University of Science

Research Assistant

Feb 2019 – Sep 2021

Ho Chi Minh City, Vietnam

- Built a Vietnamese speech corpus (19 GB) and conducted research on text normalization and phonology.
- Organized ERC2019 – Emotion Recognition Challenge.
- Teaching assistant at VNSigma Python beginner class, supported by the American Consulate.

#### Robotics & IoT Lab, University of Science

Research Assistant

May 2019 – Sep 2021

Ho Chi Minh City, Vietnam

- Teacher for EV3 Mindstorm and coach for First Lego League (FLL), EVJ Makethon, and World Robot Olympiad (WRO) competitions.

#### Computational Linguistics Center, University of Science

Research Assistant

Apr 2019 – Sep 2019

Ho Chi Minh City, Vietnam

- Teaching assistant for the Faculty of English Linguistics (USSH), guiding students in SDL Trados Studio, a computer-assisted translation software.

### PUBLICATIONS

- **APSIPA ASC 2026:** ViInstruct: A Large-Scale Vietnamese Dataset for Instruction-Based Text-to-Speech Synthesis.
- **APSIPA ASC 2026:** ViNonverbal: The First Vietnamese Nonverbal Speech Dataset via Scalable Pseudo-Labeling.
- **APSIPA ASC 2026:** Vi-SparkRL: Enhancing Tonal Accuracy and Naturalness in Vietnamese TTS via Multi-Objective Rewards.
- **MAPR 2025:** Unified Acoustic Representation Learning for Vietnamese Speech Classification Tasks.
- **MAPR 2025:** Analyzing the Correlation and Impact of Speech Evaluation Metrics on Speaker Verification and ASR.
- **MAPR 2025:** Adapting WavLM for Vietnamese Speaker Diarization in Real-world Conversations.
- **RIVF 2024:** Enhancing Deepfake Detection: WavLM and Advanced RawBoost Augmentation.
- **Voice Privacy 2024:** Voice Attacker: Multi-Head Factorized Attentive Reconstructor and Gradient Reversal for Random Prosody Anonymization.
- **DCASE 2023:** Sound Event Detection with Soft Labels Using Self-Attention for Global Scene Feature Extraction.
- **International Journal of Asian Language Processing:** FastSpeechStyle: Fast, Emotion-Controllable, High-Quality

Speech Synthesis.

- **DCAI/NeurIPS 2021:** Vietnamese Speech-based Question Answering over Car Manuals.
- **7th NAFOSTED 2020:** Vietnamese Speech Synthesis with End-to-End Model and Text Normalization.
- **MediaEval 2020:** Emotion Classification Using WaveNet Features with SpecAugment and EfficientNet.

**PyPi package** vinorm: Vietnamese Text Normalization (34,933+ downloads / 6 months).

**PyPi package** viphoneme: Vietnamese Grapheme-to-IPA Phonetization (4,195+ downloads / 6 months).

**Thesis:** DeepSpeechVC – Voice Cloning Framework with Speech Synthesis and Voice Conversion.

## AWARDS & RECOGNITION

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|--------------------------------------------------------------------------|------|
| • Outstanding Employee of the Year – VinBigdata                          | 2023 |
| • First Place, VLSP TTS Emotional Speech Synthesis                       | 2022 |
| • First Prize, Science-A-Thon, 9th Vietnamese Summer School of Science   | 2022 |
| • Outstanding Student Award in Artificial Intelligence, Ho Chi Minh City | 2021 |
| • Excellent Thesis – Science and Technology Student Award                | 2021 |
| • Runner-Up, KMS Hackathon                                               | 2020 |

## CERTIFICATIONS & COMPETITIONS

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|------------------------------------------------------------------------------------------|------|
| • Contributed to obtaining ISO/IEC 19795-1:2021 (NIST/NVLAP) for Voice Biometric Testing | 2024 |
| • Participation, IEEE Spoken Language Technology Workshop Hackathon                      | 2022 |
| • 7th/23 teams, IEEE Signal Processing Society's Signal Processing Cup                   | 2022 |
| • FPT Scholarship, Final Round, Code War Competition                                     | 2019 |
| • Final Round, Samsung Collegiate Programming Cup                                        | 2018 |
| • ACM-ICPC Vietnam National Programming Contest – Rank 33                                | 2018 |
| • Self-Driving Cars Specialisation – University of Toronto (Coursera)                    | 2018 |

## VOLUNTEERING

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| • Volunteer for medical research on dental surgery at HCMC Oromaxillofacial Hospital | 2021 |
| • F1 contact tracing support during the Covid-19 pandemic                            | 2022 |
| • Organized ERC2019 – Emotion Recognition Challenge                                  | 2019 |
| • Teaching assistant at VNsigma Python beginner class (American Consulate)           | 2019 |